LEGAL RESEARCH AND EVIDENCE ASSISTANT - JURIAID

Project ID: 25-26J-274

PROJECT PROPOSAL REPORT

Kaushal R.K.A.C

BSc. (Hons) Degree in Information Technology

Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

July 2025

JuriAid: An AI-Powered Legal Assistant System for Automated Question Generation in Legal Practice

BSc. (Hons) Degree in Information Technology Specializing in

Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology, Sri Lanka

July 2025

Author Details,

1. Author Name – Kaushal R.K.A.C
2. Student ID No. – IT22893802
3. Research objective – Agentic AI Framework Orchestrator

Supervisor Details,

1. Supervisor Name – Mr. Jeewaka Perera
2. Co-Supervisor Name – Ms. Madusha Weerasooriya
3. External Supervisor – Mrs. Nadeeka Senavirathne

Group Details,

Member Name	Student ID
Sammani W.M.W.L.	IT22318848
Kaushal R.K.A.C.	IT22893802
Rasindu N.H.A.	IT22092410
Prarthana A.P.S.	IT22127228

Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

Name	Student ID	Signature
Kaushal RKAC	IT22893802	

Statement of the supervisor,

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor:

Date: 29-08-2025.

Signature of the co-supervisor:

Date: 29/08/2025

Abstract

Legal help in Sri Lanka isn't available where people need it most. You'll find experienced lawyers concentrated in Colombo and a few other major cities, but if you live in a rural area, good luck finding quality legal assistance. Meanwhile, the lawyers who do practice still waste countless hours on manual research - flipping through books, searching physical files, doing things the same slow way they did decades ago. Here's what we noticed: nobody has built a smart system that can actually handle the messy, complicated work that legal cases require while managing different AI tools designed specifically for how Sri Lankan law works. So, we decided to build one ourselves. We're using something called LangChain to create what's basically a digital legal coordinator. Picture this: someone comes to you with a legal problem. Instead of spending hours doing everything yourself, this system takes that problem, chops it up into pieces that make sense, and hands each piece to a different specialist. One specialist digs through old court cases, another figures out which laws apply, and a third one asks smart questions to fill in the gaps. We built in a memory system so it remembers what happened in previous conversations, plus a thinking component that mimics how seasoned lawyers actually work through problems. When everything runs smoothly together, you get comprehensive legal analysis that actually fits the situation. Why does this matter? We're not just tweaking something that already exists - we're building something brand new that can do the heavy lifting legal experts normally do by hand. Practicing lawyers should be able to work faster and prepare better cases. More importantly, regular people across Sri Lanka might finally be able to get decent legal advice without traveling to the capital.

Keywords: Legal AI, Multi-agent Systems, Legal Accessibility, Sri Lankan Jurisprudence, LangChain Framework

TABLE OF CONTENTS

Abstract ..... **iv**

LIST OF FIGURES ..... **vi**

LIST OF TABLES ..... **vi**

LIST OF ABBREVIATIONS ..... **vi**

LIST OF APPENDICES ..... **vi**

1. Introduction **1**

1.1. Background & Literature survey 2

1.2. Research Gaps 4

1.3. Research Problems 6

2. Objectives **7**

2.1. Main Objectives 7

2.2. Specific Objective 7

3. Methodology **8**

3.1. Overall System 8

3.2. Individual Component Diagram (Agentic AI Framework) 9

3.3. Technology Stack 10

4. Project Requirements **11**

4.1. Functional Requirements 11

4.2. Non-Functional Requirements 11

5. Budget **13**

6. Commercialization **14**

7. Grant chart **15**

References **16**

APPENDICES **18**

LIST OF FIGURES

Figure 1:System Overview Diagram 8
Figure 2 : Component system diagram 9
Figure 3 : Component technology diagram 10
Figure 4 : Grant chart 15

LIST OF TABLES

Table 1 Comparative analysis highlighting the novelty of the proposed system. 5
Table 2: Budget Table 13

LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
LLM	Large Language Models
MoE	Mixture-of-Experts
NLP	Natural Language Processing
RMA	Reflective Multi-Agent
SOP	Standard Operating Procedure

LIST OF APPENDICES

Appendix A : Turnitin Similarity Score 18

1. Introduction

Sri Lanka's justice system has a serious problem that most people don't talk about enough - there's a huge gap between who needs legal help and who can actually get it. Here's what's happening: more than 70% of all practicing lawyers work in Colombo and a few other big cities [1]. If you live anywhere else in the country, finding a decent lawyer becomes incredibly difficult. And even when lawyers are available, they're still doing research the same way people did it thirty years ago - manually searching through books and basic digital databases that can't really understand what they're looking for [2].

We kept asking ourselves: why hasn't anyone built something better? That's where the 'JuriAid' project comes in. Our approach moves beyond simply scanning documents and calling it "digital transformation." We aim to develop a system capable of reasoning through legal issues in a manner similar to seasoned legal practitioners. This technology is intended not as a replacement for lawyers, but as a supportive tool designed to manage laborious preliminary work, thereby freeing up legal professionals to focus on strategic analysis [3].

However, building a functional AI system for the legal domain requires more than just bundling different AI tools together and hoping they cooperate. A central intelligence must be in place to direct their actions. This is the precise function of our 'Agentic Framework.' This framework essentially acts as the senior partner in a law firm. It's responsible for analyzing a complex case, delegating the various parts to the right specialists, and then weaving their work into a single, cohesive strategy [4].

The success of the whole system hinges on how well this coordination works. It wouldn't matter if we had the most sophisticated individual AI components available; if they cannot cooperate effectively, the system as a whole would be useless. This research project stems from a key insight: getting AI tools to function as a team in the Sri Lankan legal system, with all its unique quirks, demands a smart and capable coordinator. Therefore, our work involves building and testing this LangChain-based Agent**ic AI** Framework to confirm its ability to manage legal AI systems in complex scenarios [5].

1.1. Background & Literature survey

How lawyers use artificial intelligence is going through a massive change. We've moved past simple data storage and search and are now focused on building smarter systems that can act independently. This change comes from new autonomous agents built on Large Language Models (LLMs). The purpose of these agents is to observe their surroundings, think, and then act to meet a goal [6]. What makes these "agentic" systems different from older AI is their ability to plan, recall information, and use tools-skills that are a perfect match for the complex, step-by-step logic of legal work [7].

These agents generally have four key parts: profiling, memory, planning, and action [6]. The Profiling component gives an agent a specific role or "persona" that shapes its behavior, a core idea in legal AI. For instance, systems like Chatlaw and LawLuo mimic a law firm's structure by creating agents for roles like 'Legal Researcher' or 'Senior Lawyer' [8], [9]. The Memory component is essential for keeping track of long and complex legal cases. This includes short-term memory (what the LLM is currently processing) and long-term memory, which is often handled by external tools like Redis [6], [10].

The Planning function is the agent's strategic core, breaking down large tasks into smaller, more manageable steps. The Action function is what allows the agent to interact with its environment, for example, by using an API to retrieve data from a legal database or knowledge graph [11], [9].

A major downside of LLMs is their habit of "hallucinating," generating information that appears credible but is actually false-a problem that multi-agent systems aim to tackle [8], [12]. The danger of these hallucinations became very real when a lawyer submitted a brief with fake cases made up by ChatGPT [13]. For this reason, the work on multi-agent systems is vital; it's less about a tech upgrade and more about building tools people can trust to be safe and more reliable. In response to these issues, researchers have created several key frameworks.

To address these challenges, researchers have developed several notable frameworks. Chatlaw functions as a virtual legal assistant that adheres to a strict set of operating procedures, much like a real law firm. It uses a Mixture-of-Experts (MoE) model and leans on knowledge graphs to make sure the information it provides is accurate [8]. LawLuo also mimics a law firm's structure, but its

focus is on offering personalized legal help over a series of conversations. Its most distinctive feature is a "Boss" agent that reviews the work of other agents, offering a practical way to incorporate feedback into the planning process [9]. PAKTON is a specialized tool designed for analyzing lengthy legal agreements. Its main technical contribution is a more sophisticated document processing method that uses Hierarchical Parsing and Context-Aware Chunking to maintain the logical integrity of a contract during information retrieval [14]. Finally, the Reflective Multi-Agent (RMA) [15] framework offers a different model focused on internal quality assurance. It uses two specialized agents for review: a Factor Analyst to verify facts and an Argument Polisher to refine the quality of the legal arguments.

Looking at these systems, two different approaches become clear. Some, like Chatlaw and LawLuo, try to copy how a law firm works ("Process Emulation"). Others, like PAKTON and the RMA framework, focus on copying how a lawyer thinks ("Cognitive Emulation"), including the process of questioning and self-correction.

1.2. Research Gaps

After diving deep into everything that's been published about LLM-based autonomous agents in legal work, we've spotted two major holes in the research that nobody seems to be talking about much. These gaps are exactly what we're trying to tackle with the JuriAid Agentic AI Framework.

The first problem is technical. All these impressive systems we've been reading about are great at what they do, but they're all missing something crucial. Most of them just follow a basic script: first do this, then do that. ChatLaw, for example, has what they call a Standard Operating Procedure (SOP) that's basically just a flowchart telling each agent when it's their turn to work [8]. But real legal work doesn't follow flowcharts [16]. What we really need is something like a senior partner in a law firm - someone who can look at a legal problem, break it down intelligently, and then coordinate all the junior associates to work on the right things in the right order, changing the plan when new information comes to light. None of the existing systems can do this kind of real-time strategic coordination [6], [7].

The second gap is contextual. Almost everything published in this field focuses on either American/British contract law or Chinese legal systems [7], [8], [9]. But what about everywhere else? Sri Lanka is a perfect example. We have our own statutes, our own court precedents, our own legal procedures. Nobody has seriously tried to build advanced AI legal systems for contexts like ours. The technical challenges are completely different when you're working with limited data and a unique legal system [17]. But the potential impact is huge because these are exactly the places where AI could make the biggest difference in people's lives.

So we're basically trying to solve two problems at once: building smarter agent coordination technology, and proving that this technology can work in legal systems that have been completely ignored by the research community so far.

Table 1 Comparative analysis highlighting the novelty of the proposed system.

Framework	Architecture Type	Key Components	Memory System	Strengths	Limitations	Sri Lankan Law Compatibility
ChatLaw [8]	Process Emulation (MoE)	Legal Researcher, Senior Lawyer agents	Knowledge Graph + Short-term	Structured SOP workflow, Fact verification	Static workflow, Limited adaptability	Low - US/Chinese law focused
LawLuo [9]	Hierarchical Multi-agent	Boss Agent, Specialist agents	Conversation buffer	Personalized consultations, Feedback loops	Linear processing, No long-term memory	Medium - Adaptable structure
PAKTON [14]	Document-centric	Hierarchical Parser, Context Analyzer	Document-based chunking	Advanced contract analysis	Limited to document analysis	Medium - Document processing applicable
RMA Framework [15]	Cognitive Emulation	Factor Analyst, Argument Polisher	Internal review system	Quality assurance, Self-correction	Narrow scope, No case continuity	Low - Specialized for arguments only
JuriAid (Proposed)	Dynamic Orchestration	Agentic AI Framework, Memory Manager, Reasoning Engine	Redis + ConversationBuffer	Adaptive planning, Sri Lankan law focus, Long-term memory	Under development	High - Designed specifically for Sri Lankan jurisprudence

1.3. Research Problems

Our review of the current research on multi-agent systems for legal AI has led us to a core question that remains largely unanswered:

How can we build a central coordination system that manages multiple specialized AI agents for legal work, keeps track of complex conversations, and mimics how experienced Sri Lankan lawyer's approach legal problems?

This question is the foundation of our research. We aim to do more than just connect different AI tools; we want to build a system that can manage a legal consultation from the initial client meeting to the final advice. Our research is designed to tackle four specific technical hurdles:

Smart Task Management: Most legal AI systems currently work in a very linear fashion, a direct result of their static workflow designs [8]. Our system needs to make strategic decisions and be flexible enough to change course when the initial analysis uncovers something unexpected.

Conversation History Management: Legal consultations are like extended detective work. Most AI systems treat each question as if it exists in a vacuum, a known challenge in autonomous agents [6]. We need something that can follow the thread of a long, complicated legal conversation.

Replicating Legal Thinking: How do you teach a computer to think like a lawyer with twenty years of experience? Human lawyers develop this kind of strategic thinking through years of practice [16]. We need to figure out how to build that same adaptability into our AI system.

Adapting to Sri Lankan Law: Almost all current legal AI has been developed for the American or Chinese legal systems [7], [9]. An AI trained for a law firm in New York simply won't work in a courtroom in Colombo. A key part of our research, therefore, is to find ways to adapt these technologies for a legal system that has been completely ignored by the major players in the field.

Solving these problems would result in a system with the potential to transform legal services in Sri Lanka and offer a valuable contribution to the broader field of legal AI.

2. Objectives

2.1. Main Objectives

This research is all about building and testing the "Agentic AI Framework." We want it to act like a smart manager for JuriAid's AI tools, thinking through problems the way an expert lawyer would [16]. The end goal is a tool that gives reliable legal advice that gets the full picture of a case under Sri Lankan law.

2.2. Specific Objective

- **Create a central coordinating system with LangChain.** This will be the project's engine, taking a user's query, breaking it into logical tasks, and sending each task to the right specialist agent (like the Court Retriever, Law Retriever, or Questionnaire Generator) [5].

- **Implement a memory system for ongoing conversations.** To keep a long-term record of each case, we'll use a permanent data store like Redis [11]. This will allow the system to follow extended discussions and manage complex cases that evolve over time without losing important details [6], [10].

- **Create a reasoning engine using a mix of rules and model-based logic.** That engine will be the system's "mind." It will analyze the user's question and the facts of the case to select the right agent for the task, giving the framework its intelligence [6].

- **Use FastAPI to create a reliable API for communication between all system parts.** This will be done by creating a standard set of REST APIs to connect the main coordinator with the specialist agents, which will make the system modular and easy to build on in the future.

- **we will establish a clear plan for handling errors.** Our plan will involve using retries, timeouts, and backup strategies to keep the system running without any interruption, even when things go wrong.

3. Methodology

3.1. Overall System

We're going to build the Agentic AI Framework using a step-by-step process that combines well-known ideas from autonomous agent design [6], [7] with modern software development methods. To really get what the Agentic AI Framework does, it helps to see where it fits into the bigger JuriAid system. It's essentially the project manager for a team of specialized AI tools, a concept explored in multi-agent conversation frameworks [4].

Here’s how it works: When a user asks a legal question, the Agentic AI Framework Orchestrator takes charge. It then uses three other specialist agents to get the job done: one that finds past court decisions, one that looks up laws, and one that generates clarifying questions to ask the user. This multi-agent workflow is similar to architectures seen in other legal AI systems [8], [9].

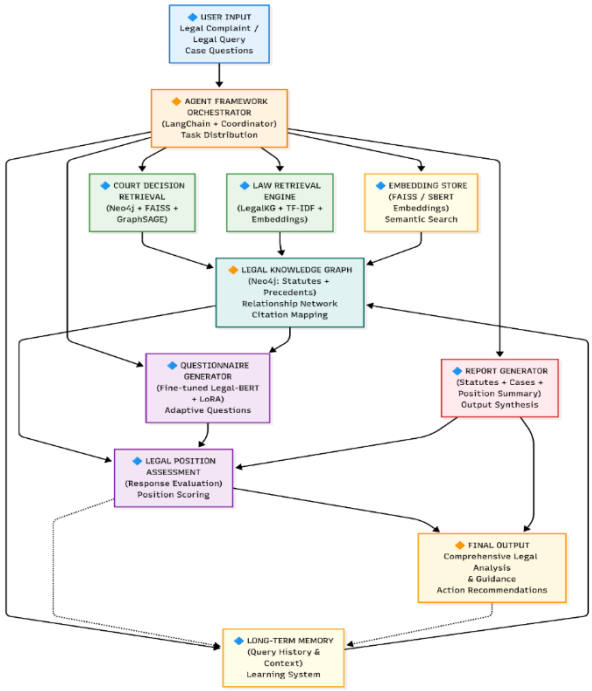

Figure 1:System Overview Diagram

3.2. Individual Component Diagram (Agentic AI Framework)

The core of this research project is the detailed design and implementation of the Agentic AI Framework itself, structured around the unified framework for LLM-based autonomous agents [6]. The implementation will directly map the theoretical modules of an autonomous agent to concrete technologies. The "profiling" of agents is achieved through their explicit, specialized roles. The "planning" module, the central intelligence, will be implemented using LangChain's agent constructs to generate a dynamic, multi-step plan for each user query [5]. The "memory" module will be a hybrid system, using LangChain's “ConversationBufferWindowMemory” for short-term context and a Redis-backed “ConversationBufferMemory” for long-term case continuity [10], [11]. The "action" module is the framework's ability to execute its plan by using the specialist AI agents as "tools," interacting with them through an API management layer [6].

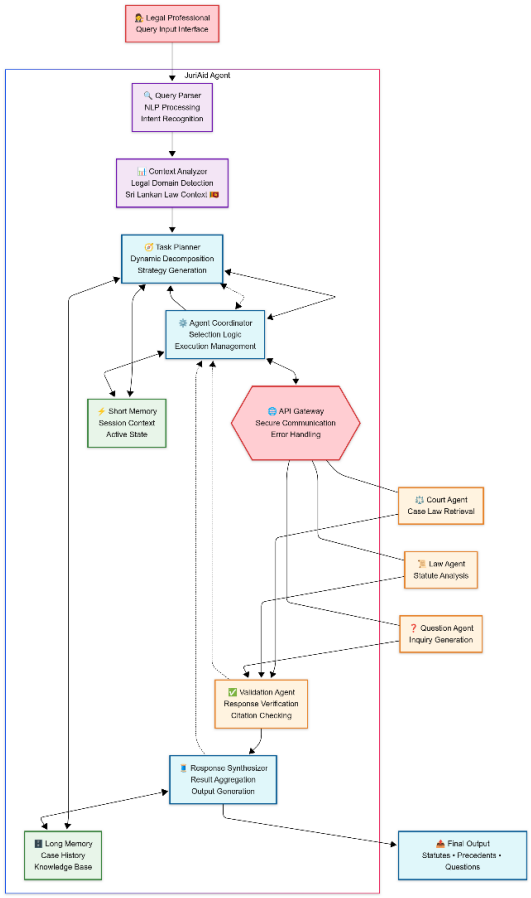

Figure 2 : Component system diagram

3.3. Technology Stack

We've chosen our technology based on tools that are well-established and trusted in the field. The agent's core thinking will be built with LangChain [5] and LlamaIndex [18]. We'll use FastAPI for the API, Redis [11] for long-term memory, and Python for all the coding. This tech combination is designed to handle the entire workflow of a legal question—from understanding the initial query to finding the right documents and asking follow-up questions—all directed by the central orchestrator.

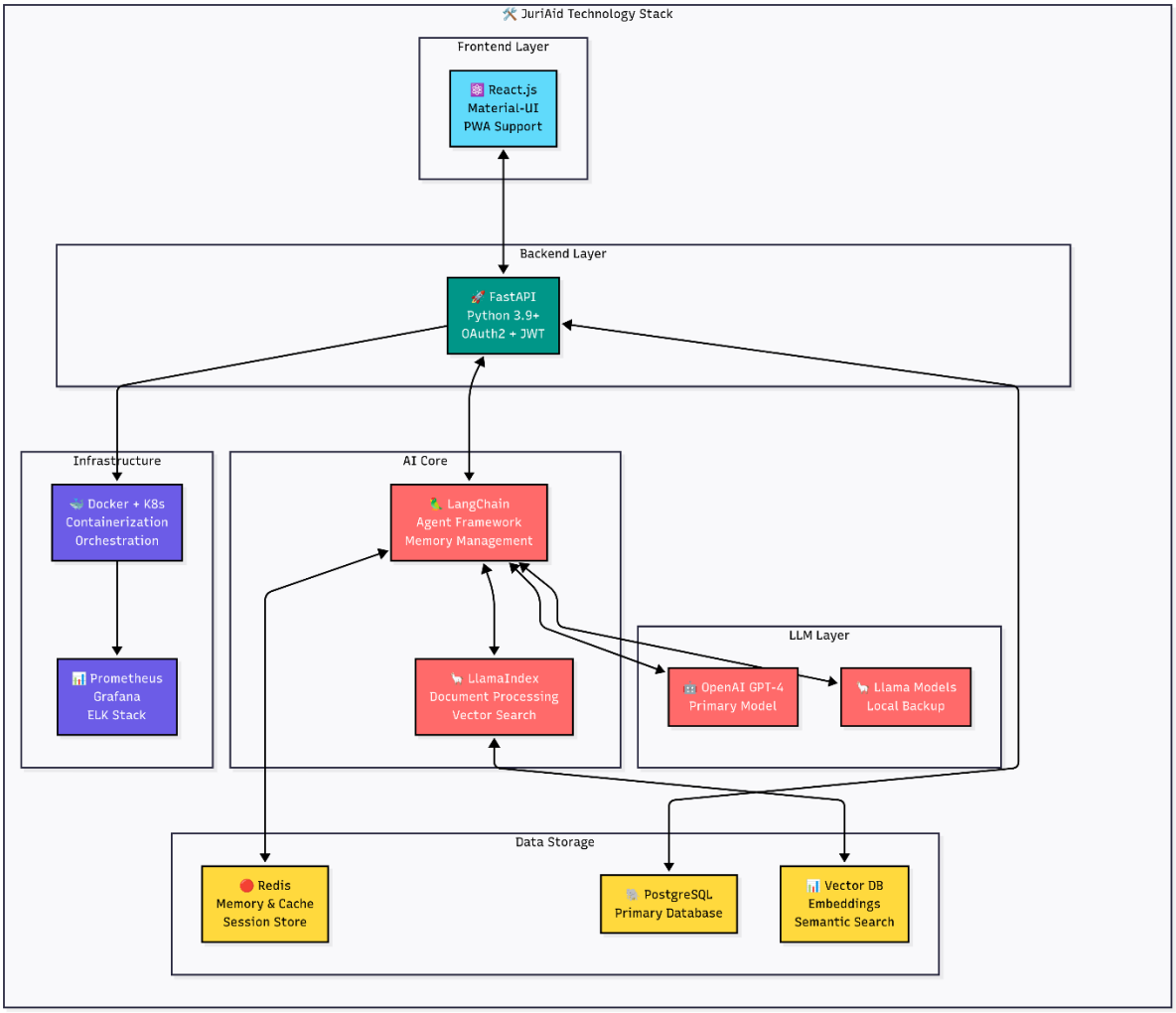

Figure 3 : Component technology diagram

4. Project Requirements

4.1.Functional Requirements

Functional requirements outline the precise actions, duties, or responsibilities that the system must carry out. For the Agentic AI Framework component, these include:

- **Query Processing:** The system shall accept and correctly interpret legal queries submitted in natural language English.
- **Task Decomposition:** The system shall analyze the user's query and autonomously decompose it into a logical sequence of executable sub-tasks.
- **Intelligent Agent Delegation:** The system shall accurately identify and delegate each sub-task to the appropriate specialist AI agent via a REST API call.
- **Stateful Context Management:** The system shall maintain the complete conversational history and contextual state for each unique legal case.
- **Response Synthesis:** The system shall be capable of aggregating the individual responses from all specialist agents and synthesizing them into a single, coherent final output.
- **Error Management:** The system shall detect and handle errors or timeouts from subordinate agents.

4.2.Non-Functional Requirements

Non-functional requirements define the criteria that describe the system's behavior and quality attributes.

- **Security:** All data in transit and at rest must be encrypted, complying with Sri Lanka's Personal Data Protection Act.
- **Reliability:** The Agentic AI Framework must be highly reliable, with a target uptime of 99.9%, and resilient to failures in subordinate agents.
- **Efficiency:** The end-to-end processing time for a standard legal query should not exceed 60 seconds.

- **Scalability:** The architecture must be designed for horizontal scalability and be deployable on cloud infrastructure.
- **Usability:** The API design must be clear, consistent, and well-documented to ensure easy integration for other team members.

5. Budget

The following amounts are estimated and may be subject to change:

Table 2: Budget Table

Description	Cost	Occurring
Server Hosting	Covered by GitHub Pack	Monthly
Cloud Database (e.g., AWS RDS for legal data storage)	Covered by GitHub Pack	Monthly
Legal Metirials	LKR 10,000	Annual
Developer Tools/Software Subscriptions	LKR 15,000 (Estimated Value)	Annual
LLM API Usage	LKR 12,000	Monthly

6. Commercialization

It is advised to launch this system as a **stand-alone entity** rather than being bound to a cooperative organization, especially given the current lack of a centralized, efficient platform in Sri Lanka that connects legal professionals with such AI tools. This approach ensures that any lawyer, regardless of their firm or affiliation, can utilize the system.

The system will be designed for **multi-device support**, with a dedicated **Mobile application for lawyers** to access its features. It will likely adopt a **freemium business model**. A free tier could offer basic question generation functionalities or a limited number of questions per case. The premium tier would unlock advanced features crucial for comprehensive legal practice, such as:

• **Unlimited, sophisticated question generation** with deeper legal context and nuance.

• **Integration with broader legal knowledge bases** and real-time legal updates.

• **Advanced fact elicitation and iterative refinement capabilities.**

7. Grant chart

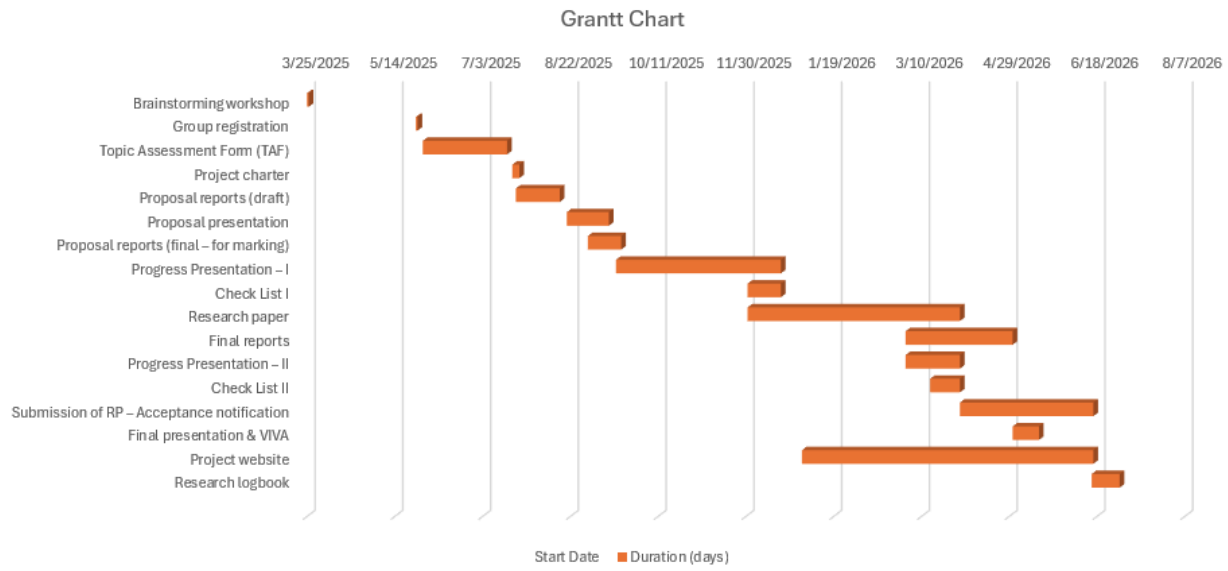

Figure 4 : Grant chart

References

[1] D. Mendis, "The Digital Transformation of Legal Services in Sri Lanka: Challenges and Opportunities," *Bar Association of Sri Lanka Law Journal*, vol. 25, no. 1, pp. 45-62, 2022.

[2] H. Mentzingen, A. António, F. Bacao and S. Prakash, "Automation of Legal Precedents Retrieval: Findings from a Literature Review," *International Journal of Intelligent Systems*, p. 6660983, 2023.

[3] R. Susskind and D. Susskind, The Future of the Professions: How Technology Will Transform the Work of Human Experts, Oxford, UK: Oxford University Press, 2015.

[4] Q. Wu, G. Bansal, J. Zhang, Y. Wu, B. Li, E. Zhu, L. Jiang, X. Zhang, S. Zhang, J. Liu, A. Awadallah, R. W. White, R. W. White, D. Burger and C. Wang, "AutoGen: Enabling next-gen llm applications via multi-agent conversation," 2023.

[5] LangChain, "LangChain Documentation," 2024. [Online]. Available: https://python.langchain.com/docs/. [Accessed 13 August 2025].

[6] L. Wang, C. Ma, X. Feng, Z. Zhang, H. Yang, J. Zhang, Z. Chen, Tang, X. Chen, Y. Lin, W. X. Zhao, Z. Wei and J.-R. Wen, "A Survey on Large Language Model based Autonomous Agents," *Frontiers of Computer Science*, vol. 18, no. 6, p. 186345, December 2024.

[7] S. Franklin and A. Graesser, "Is It an agent, or just a program?: A taxonomy for autonomous agents," in *Intelligent Agents III Agent Theories, Architectures, and Languages*, vol. 1193, J. P. Müller, M. J. Wooldridge and N. R. Jennings, Eds., Berlin, Heidelberg, Springer Berlin Heidelberg, 1997, pp. 21-35.

[8] J. Cui, M. Ning, Z. Li, B. Chen, Y. Yan, H. Li, B. Ling, Y. Tian and L. Yuan, "Chatlaw: A Multi-Agent Collaborative Legal Assistant with Knowledge Graph Enhanced Mixture-of-Experts Large Language Model," 2024.

[9] J. Sun, C. Dai, Z. Luo, Y. Chang and Y. Li, "LawLuo: A Multi-Agent Collaborative Framework for Multi-Round Chinese Legal Consultation," 2024.

[10] LangChain, "Redis Integration," 2024. [Online]. Available: https://python.langchain.com/docs/integrations/providers/redis/. [Accessed 11 August 2025].

[11] Redis, "Redis Documentation," 2024. [Online]. Available: https://redis.io/docs/latest/. [Accessed 9 August 2025].

[12] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto and P. Fung, "Survey of Hallucination in Natural Language Generation," *ACM Computing Surveys*, vol. 55, no. 12, pp. 1-38, 2023.

[13] B. Weiser, "A.I. Shows Its Mischievous Side: Making Up Fake Cases," *The New York Times*, 27 May 2023.

[14] P. Raptopoulos, G. Filandrianos, M. Lymperaiou and G. Stamou, "PAKTON: A Multi-Agent Framework for Question Answering in Long Legal Agreements," 2024.

[15] L. Zhang and K. D. Ashley, "Mitigating Manipulation and Enhancing Persuasion: A Reflective Multi-Agent Approach for Legal Argument Generation," 2025.

[16] M. A. Eisenberg, "Legal Reasoning," in *The Blackwell Guide to the Philosophy of Law and Legal Theory*, M. P. Golding and W. A. Edmundson, Eds., Malden, MA, Blackwell Publishing, 2005, pp. 46-61.

[17] U. Kohl, "The Rule of Law in the Digital Age," in *The Cambridge Handbook of the Law of the Digital Society*, N. A. Moreham, M. R. F. S. D. and D. L., Eds., Cambridge, UK, Cambridge University Press, 2020.

[18] J. Liu, "LlamaIndex Documentation," 2024. [Online]. Available: https://docs.llamaindex.ai/en/stable/. [Accessed 18 August 2025].

APPENDICES

Turnitin Originality Report

Processed on: 27-Aug-2025 16:07 IST
ID: 2736107295
Word Count: 3943
Submitted: 1

IT22893802.docx By Chathumina Kaushal

Similarity Index	Similarity by Source	
6%	Internet Sources:	5%
	Publications:	1%
	Student Papers:	5%

4% match (student papers from 22-Mar-2021)
Submitted to Sri Lanka Institute of Information Technology on 2021-03-22

1% match (student papers from 07-Aug-2020)
Submitted to Sri Lanka Institute of Information Technology on 2020-08-07

< 1% match (Internet from 21-Mar-2024)
https://www.coursehero.com/file/227387834/PMC120-BRD-Group4docx/

< 1% match (Internet from 15-Jul-2023)
https://www.coursehero.com/file/202995683/2020-138-Kolamunnadocx/

< 1% match (Internet from 09-Mar-2022)
https://www.coursehero.com/file/96283459/2020-183-IT17123228-Individual-Proposal-Reportpdf/

< 1% match (Internet from 08-Mar-2023)
https://trimis.ec.europa.eu/sites/default/files/project/documents/20121024_181856_29401_D1.5_Non_Functional_requirements_analysis_v1.6.pdf

Appendix A : Turnitin Similarity Score